



System architecture — *one source in, every format out*

A Gen-AI content transformation platform: the pipeline runs in the browser, and each model call goes to a cloud provider or to an on-device model, by the user's choice.

LEGEND

- **Private mode** · never leaves the device
- **Cloud mode** · AI requests
- **Accounts & data** · Supabase

USER'S DEVICE

BROWSER React 19 · TypeScript 5.7 · Vite 8 · Tailwind CSS 4 · installable PWA · the whole pipeline runs here

WORKSPACE UI**UI Source & configuration**

Upload PDF · DOCX · image, paste text or a link. Pick from 10 formats or describe one; tone and audience.

Processing mode

Cloud APIs

Private

Hardware check (RAM · cores · GPU) and Ollama setup guide before Private runs.

UI Output canvas

Collects every draft; on-device drafts stream in token by token.

Compare up to 3 models side by side · refine one draft in plain words · Stop at any moment.

Export PowerPoint (.pptx), Markdown or text.

UI History · Analytics · Admin

Every draft with its format, model and word counts; usage totals.

Admin: all users and activity; disable, enable or delete accounts.

Every /workspace route sits behind an auth guard.

Document readers

PDF → text (pdf.js), DOCX → text (mammoth), TXT · CSV · JSON · MD. Scans → page images.

Image reader

Vision model (Gemini, Mistral) or on-device OCR (Tesseract.js). Always OCR in Private mode.

CLIENT-SIDE INGESTION**Link reader**

Page → text through Jina Reader. Cloud mode only.

AI ORCHESTRATION · LANGGRAPH-STYLE NODE GRAPH**N1 Ingest node**

Merges text, file, link, image

N2 Parse node

Local clean-up, no AI call

N3 Fit source

Size budget per provider

N4 Format × Model

One node per (format, model), fanned out in parallel: 2–3 per provider, 1 on-device.
tone · audience · instructions

N5 Translate

Whole text, chunk by chunk

N6 Refine router

Draft only, or draft + source

N7 Run controller

Smart generation check skips drafts you already have.
One AbortController per run: Stop keeps finished drafts.

MODEL GATEWAY · CHAT()**6 Route every call**

Retries rate limits, 2-min timeout, own-key consent prompt before shared keys.

No own key → /api/chat, shared keys

Own API key → provider directly

Private mode → Ollama on this computer



PRIVATE MODE Local engine · Ollama · qwen2.5vl:7b localhost:11434

Streamed /api/generate: drafts arrive token by token; Stop closes the stream at once.

num_ctx 4096 · temperature 0.3: fits an 8 GB laptop GPU, stays close to the source.

Zero data leaves: no server, no provider, no key. History keeps counts only.

CLOUD SERVICES**DB Supabase · accounts & data****Auth**

Email sign-in and sessions. Disabled accounts are banned here and signed out everywhere.

Postgres + RLS

profiles · activity_logs. Each user reads only their own rows; the admin reads all.

Server-checked functions: admin_list_users · admin_set_user_disabled · admin_delete_user · my_account_disabled

WEB Jina Reader

Fetches a linked page server-side and returns it as text, because models can't open links themselves.

API Vercel**Static hosting**

The React app, PWA manifest and service worker.

/api/chat function

Holds the shared provider keys in environment variables; rotates them and retries rate limits.

AI Cloud AI providers

11 drafting models

Groq

gpt-oss-120b (default) · gpt-oss-20b · qwen3.8-27b. Fastest drafts; the refine router.

Gemini

3.5 Flash-Lite · 3.8 Flash. Reads text and images; the longest sources.

Mistral

Large · Medium · Small · Nemo · Codestral. Vision on Medium and Small.

Users may paste their own key; it stays in the tab's memory and is sent only to that provider.

PRIVATE

In Private mode none of this column sees the document: no text, images, drafts or file names reach the cloud.

10 formats

+ any custom format, drafted in parallel

3 + 1 engines

Groq · Gemini · Mistral, or Ollama on-device

4,096 tokens

local window, sized for an 8 GB GPU

6 languages

full-length Indian-language translation

RLS on every table

data access enforced by the database

How the architecture works

REQUEST LIFECYCLE

- 1
- Sign in.** Supabase Auth issues the session; every workspace route is behind an auth guard, and disabled accounts are refused.
- 2
- Configure.** Source, formats, tone, audience and engine, plus the processing mode: Cloud or Private.
- 3
- Read.** PDF and DOCX become text in the browser; images and scans go to a vision model, or to on-device OCR in Private mode.
- 4
- Plan.** The smart generation check sends only the formats not already on the canvas, so no tokens are spent twice.
- 5
- Draft.** One node per (format × model) runs in parallel under per-provider limits; on-device drafts stream one at a time.
- 6
- Review.** Compare models side by side, refine one draft in plain words, stop at any moment, export to PowerPoint, Markdown or text.
- 7
- Record.** One History row per model. Private runs record counts, formats and model name, never the text.

TWO PROCESSING MODES

	Cloud APIs · fast	Private · on-device
Drafts written by	Groq · Gemini · Mistral	qwen2.5vl:7b via Ollama
Document text	Sent to the provider	Never leaves the device
Images & scans	Vision model	OCR in the browser
Links	Jina Reader	Off, by design
History keeps	Excerpt and drafts	Counts only
Best for	Speed, long sources, comparing models	Legal, finance and government documents

Why this design. Keeping the pipeline in the browser means no document store to breach and no server that sees the text in Private mode. The only server code is one small function that holds the shared keys.

INGESTION NODES

Document readers documents.ts	pdf.js and mammoth turn PDF and DOCX into text in the browser; a scanned PDF's first 10 pages become images.
Image reader ingest.ts	Uses a selected vision model, else Gemini or Mistral; falls back to Tesseract.js OCR. Private mode always uses OCR, so images stay local.
Link reader readUrl	Fetches the page as text through Jina Reader, since models can't open links. Switched off in Private mode.

ORCHESTRATION NODES

N1 Ingest ingestNode	Merges pasted text, file text, link text and image text into one source.
N2 Parse parseNode	Normalises the text locally. No model call, so nothing is cut off and no quota is spent.
N3 Fit source fitSource	Sizes the source to each engine (Groq 12k, Mistral 60k, Gemini 200k, local 6k characters), sampling evenly across long documents.
N4 Format × Model formatNode	Writes one format with the tone, audience and custom instructions. Fanned out in parallel: 2–3 at a time per provider, one at a time on-device.
N5 Translate translateNode	Translates the whole document into six Indian languages, chunk by chunk, halving any chunk that runs out of room.
N6 Refine router routeRefinement	A fast model decides whether an edit needs the draft only or the source too. Skipped on-device, where every call stays local.
N7 Run controller runPipeline	Plans the run, then one AbortController stops every request at once, keeping finished drafts and freezing streamed ones.

GATEWAY & INFERENCE

chat() gateway providers.ts	Routes each call: own key → provider directly; no key → /api/chat; Private → Ollama. A failing own key pauses the run and asks before shared keys are used.
/api/chat Vercel function	Holds the shared keys in environment variables, rotates them and waits out rate limits. Keys never reach the browser.
Cloud providers Groq · Gemini · Mistral	11 drafting models for text and vision. Users can compare up to three on the same brief.
Local engine local.ts · Ollama	qwen2.5vl:7b at localhost:11434, streamed with num_ctx 4096 and temperature 0.3 so a 7B model fits an 8 GB GPU. A hardware check and setup guide run first.

DATA & SECURITY

Supabase Auth AuthContext.tsx	Email sign-in and sessions. An admin-disabled account is banned at the auth layer and signed out everywhere.
Postgres + RLS schema.sql	Users read only their own rows; the admin reads all. Every admin action is a server-checked function.
Activity log activity.ts	One row per model per run feeds History and Analytics. Private runs log counts only.
PWA shell sw.js · manifest	Installable on phones and desktops; the service worker caches only the site's own files.

TECH STACK

React 19 · TypeScript 5.7 · Vite 8 · Tailwind CSS 4 · Motion · pdf.js · mammoth · Tesseract.js · pptxgenjs · Supabase (Auth, Postgres, RLS) · Vercel Functions · Groq · Google Gemini · Mistral · Ollama

SECURITY & RELIABILITY AT A GLANCE

Keys stay server-side

Shared keys live only in /api/chat's environment. A user's own key stays in the tab's memory.

Consent before fallback

A failing own key pauses the run; shared keys are used only if the user agrees.

Stop means stop

One AbortController per run cancels every request and retry wait at once, keeping finished work.

Least privilege

RLS scopes every row to its owner; admin functions check the role on the server, not in the UI.